

Minxuan Alice Duan

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EDUCATION

Northeastern University, Khoury College of Computer Sciences	Boston, MA, US
Ph.D. in Computer Science, Advisor: Hongyang Zhang , Tina Eliassi-Rad .	Sept. 2025 - Present
University of Chicago, Department of Statistics	Chicago, IL, US
M.S. in Statistics.	Sept. 2021 - June 2023
Peking University, Yuanpei College & School of Mathematical Sciences	Beijing, China
B.S. in Statistics. <i>Excellent Graduates Award of Peking University</i> .	Sept. 2016 - June 2020

WORK EXPERIENCE

Novateur Research Solutions	Ashburn, VA, US
<i>Research Engineer - HAYSTAC Project under the IARPA program</i>	Aug. 2023 - Nov. 2024

- Led the development of multi-task deep learning models combining Bayesian principles with transformer architectures for anomaly detection in human mobility patterns, achieving an 11% $\rightarrow$ 65% increase in AUPRC, securing the highest agent accuracy score among five competing performers.
- Engineered a novel GNN model for anomaly detection, enabling mobility behavior analysis across multiple time windows, boosting predictive accuracy and reducing processing time by $\sim$ 12 hours.
- Optimized an exploratory data analysis pipeline for large-scale trajectory data, automating report generation and enhancing scalability, reducing analysis time for datasets with millions of agents.

Alibaba Group	Beijing, China
<i>Machine Learning Engineer Intern - Digital Media and Entertainment Lab</i>	June 2021 - Aug. 2021

- Developed a novel curriculum learning algorithm for Entity Alignment on dense knowledge graphs utilizing only structural information, resulting in a 2.7% improvement in alignment performance.
- Incorporated Inverse Probability Weighting for self-correction using Causal Inference, leading to a 4.15% increase in model accuracy with no extra computation cost.

Meituan	Beijing, China
<i>Machine Learning Engineer Intern - Autonomous Delivery PnC Team</i>	Sep. 2020 - June 2021

- Constructed a Reinforcement Learning agent training pipeline, increasing training efficiency by 5x and integrating Decision Transformer models for autonomous driving applications.
- Enhanced lane-change decision prediction by developing a data sampling strategy and employing HMM, PCA, and XGBoost to extract features from real-world/simulated multi-modal road data.
- Introduced a causal reinforcement learning framework for reward shaping to address parameter fine-tuning challenges, optimizing the transfer learning process in diverse driving scenarios.

RESEARCH EXPERIENCE

Booth School of Business	Chicago, IL, US
Advisor: Prof. X.Y. Han (UChicago)	Nov. 2024 - Present

- Investigated the lack of causal validity of current Sparse Autoencoders (SAEs) approaches by designing counterfactual tests to assess persistence of feature activations.
- Analyzed the “smearing” phenomenon (features cannot be linearly separated) as a result of spurious correlations in datasets, proposed counterfactual pairs to isolate causal feature dimensions.
- Explored adapting group lasso across layers to identify the most important features, extending beyond optimal-layer analysis to uncover cross-layer superposition.

International Computer Science Institute (ICSI)	Berkeley, CA, US
Advisor: Prof. Stella X. Yu (UMich)	June 2019 - Dec. 2019

- Designed a James-Stein shrinkage estimator based on Log-Normal Mixture Models for Manifold Learning in complex space $\mathbb{C}$ and theoretically proved its dominance over MLE methods.

Beijing International Center for Mathematical Research (BiCMR) Beijing, China
 Advisor: Prof. Hao Ge (PKU) Apr. 2018 - Nov. 2019

- Formulated a 6-state stochastic process model and conducted numerical experiments to elucidate the synergy mechanism between the noise enhancer and activator in HIV latency reactivation.
- Conducted theoretical proof of the relationship between the synergy and the direction of probability flux, contributing to a deeper understanding of the underlying processes.

PUBLICATIONS & PRIPRINTS

- [1] Z. Zhang, **M. Duan**, Y. Ye, H. Zhang, “*Scalable Multi-Objective and Meta Reinforcement Learning via Gradient Estimation*”, under review.
- [2] D. Li, Z. Zhang, **M. Duan**, E. Dobriban, H. Zhang, “*Efficiently Learning Branching Networks for Multitask Algorithmic Reasoning*”, under review.
- [3] Z. Zhang, **M. Duan**, H. Koutsopoulos, H. Zhang, “*Learning Multimodal Embeddings for Traffic Accident Prediction and Causal Estimation*”, under review.
- [4] **M. Duan**, Y. Qian, L. Zhao, Z. Zhou, Z. Rasheed, R. Yu, K. Shafique, “*Back to Bayesics: Uncovering Human Mobility Distributions and Anomalies with an Integrated Statistical and Neural Framework*”, 1st ACM SIGSPATIAL International Workshop on Geospatial Anomaly Detection, 2024.
- [5] X. Guo, T. Tang, **M. Duan**, L. Zhang, H. Ge, “*The Nonequilibrium Mechanism of Noise-Enhanced Drug Synergy in HIV Latency Reactivation*”, iScience, Vol. 25, Issue 6, 2022.
- [6] R. Chakraborty, Y. Xing, **M. Duan**, S.X. Yu, “*C-sure: Shrinkage Estimator and Prototype Classifier for Complex-Valued Deep Learning*”, 16th CVPR on Computer Vision and Pattern Recognition Workshops, 2020.

SKILLS

Programming Language: Python, R, Java, JavaScript, C/C++, SQL, Matlab

Tools and Frameworks: Git, Latex, PyTorch (Lightning), Tensorflow, Spark, Docker, React, AWS